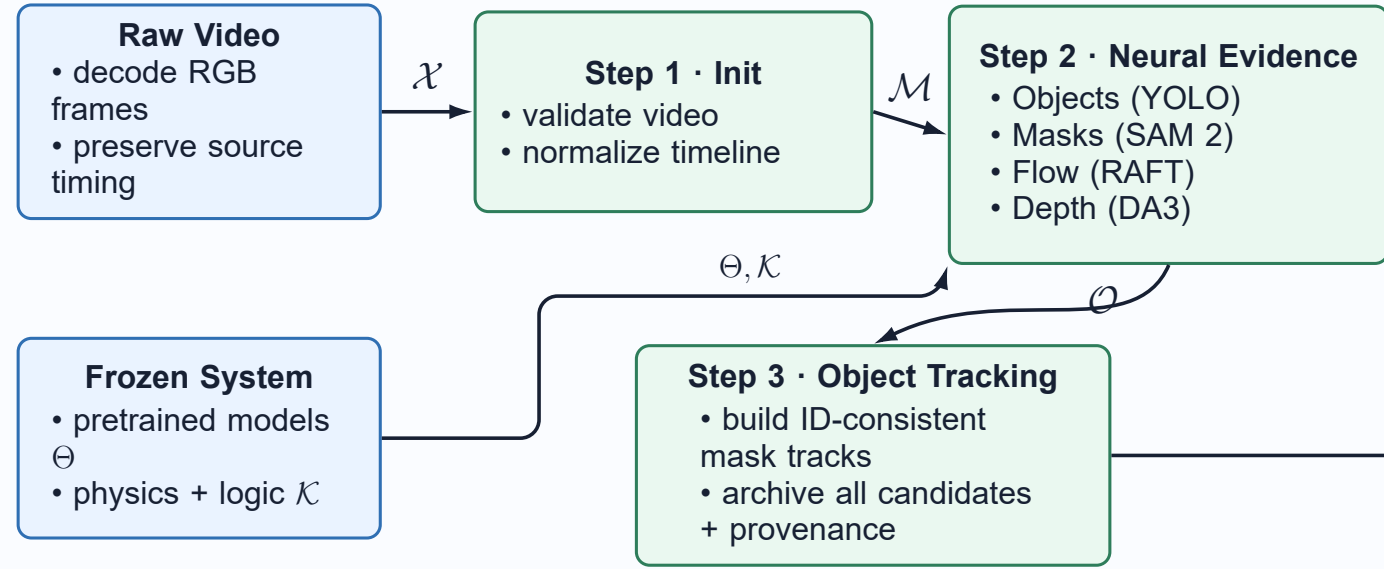


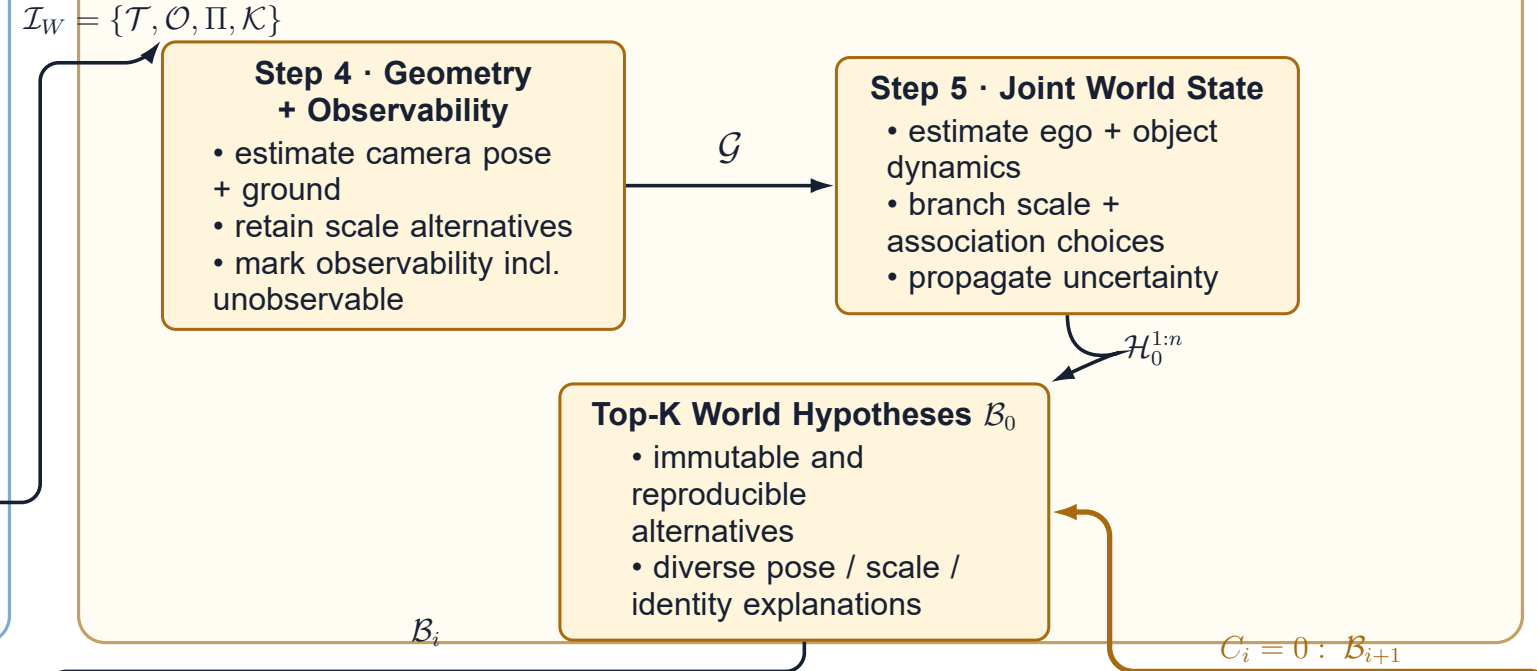
# Training-free Evidence-grounded World-State Inference

Per-video inference only · no cross-video learning · annotations remain sealed until freeze

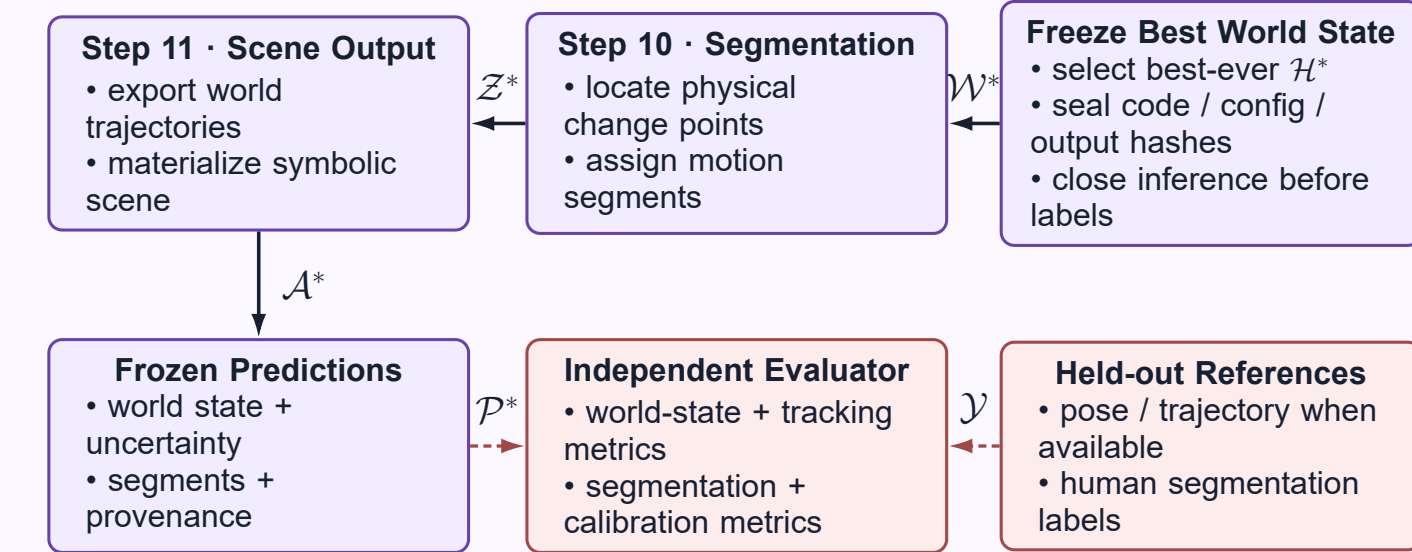
## I · VIDEO → REPLAYABLE EVIDENCE



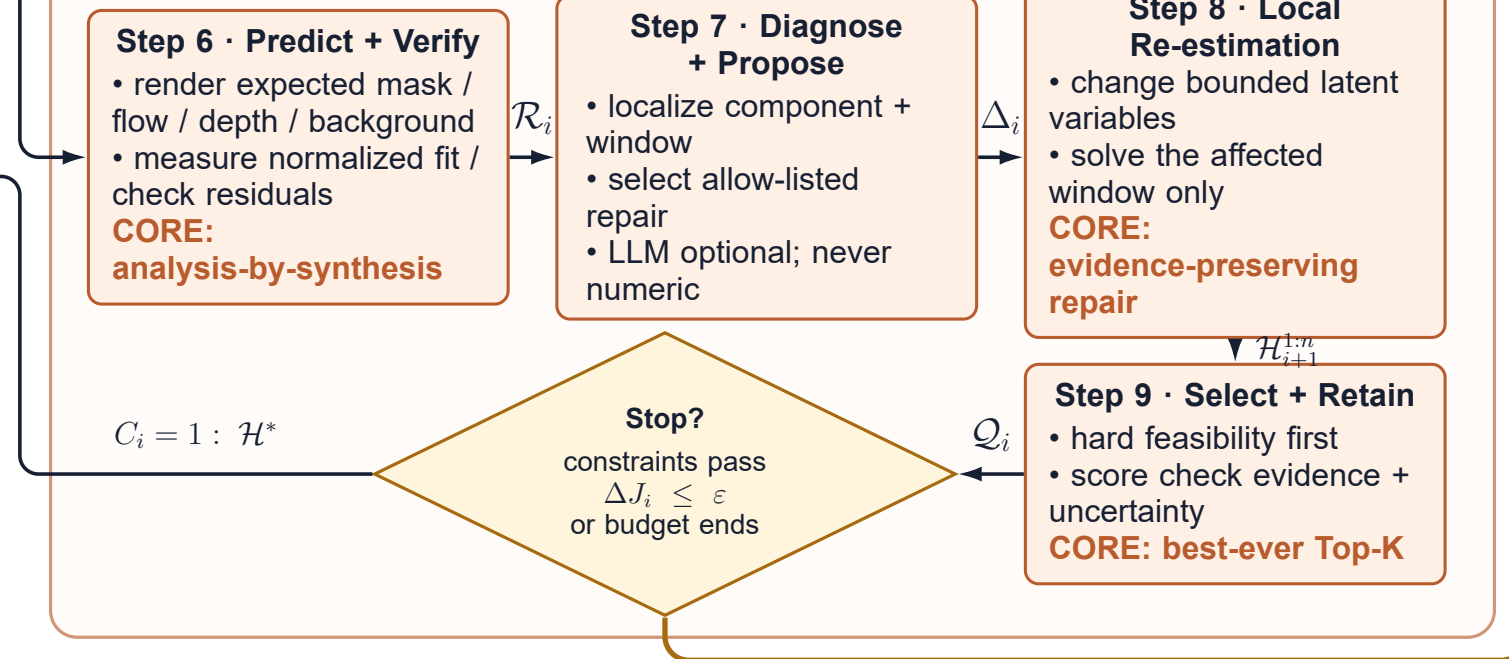
## II · UNCERTAIN WORLD HYPOTHESES



## IV · FREEZE → BLIND EVALUATION



## III · ANALYSIS-BY-SYNTHESIS LOOP



$\mathcal{X}$  video ·  $\Theta$  frozen neural models ·  $\mathcal{K}$  frozen knowledge ·  $\mathcal{O}$  independent evidence ·  $\mathcal{T}$  tracks + archive ·  $\mathcal{G}$  geometry

$\mathcal{B}_i$  world-state beam ·  $\Pi$  fit/check roles ·  $\mathcal{R}_i$  residuals ·  $\Delta_i$  bounded repair ·  $\mathcal{H}^*$  best explanation ·  $\mathcal{W}^*$  frozen world ·  $\mathcal{Y}$  evaluation-only references